EECS 16A: Introduction to Signals, Dynamics, and Information Exam 3
Department of Electrical Engineering and Computer Sciences
University of California, Berkeley 15 May 2026

FIRST Name: Einglar LAST Name: Viktor SID (All Digits): 0123456789

- **(5 Points)** On *every* page, print legibly your name and ALL digits of your SID. For every page on which you do not write your name and SID, you forfeit a point, up to the maximum 5 points.
- **(10 Points) (Pledge of Academic Integrity)** Hand-copy, sign, and date the single-line text (which begins with *I have read, . . .*) of the Pledge of Academic Integrity on page 3 of this document. Your solutions will *not* be evaluated without this.
- **Urgent Contact with the Teaching Staff:** For an urgent matter, raise your hand.
- **This document consists of pages numbered 1 through 16.** Verify that your copy of the exam is free of anomalies, and contains all of the specified number of pages. If you find a defect in your copy, contact the teaching staff immediately.
- You have 120 minutes total—*in one sitting*—to tackle this exam.
- **This exam is closed book.** You may not use or access, or cause to be used or accessed, any reference in print or electronic form at any time during the exam, except three double-sided 8.5"×11" sheets of handwritten, original notes having no appendage.

  Collaboration is not permitted.

  Computing, communication, and other electronic devices (except dedicated timekeepers) must be turned off.

  We'll provide you with scratch paper. Ask for more if needed. Do not use your own.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it.*
- For each problem, limit your work to the space provided specifically for that problem. *No other work will be considered. We will not evaluate scratch work. No exceptions.*
- Unless explicitly waived by the specific wording of a problem, you must explain your responses (and reasoning) succinctly, but clearly and convincingly.
- In some parts of a problem, we may ask you to establish a certain result—e.g., "show this" or "prove that." Even if you're unable to establish the result asked of you, you may still take that result for granted and use it in subsequent parts of the problem.
- If we ask you to provide a "reasonably simple expression" for something, by default we expect your expression to be in closed form—one *not* involving a sum $\sum$ or an integral $\int$—*unless* we explicitly tell you otherwise.
- Noncompliance with these or other instructions from the teaching staff—*including, for example, commencing work prematurely, or continuing it beyond the allocated time window*—is a serious violation of the Code of Student Conduct.
- We hope you do a *fantastic* job on this exam.

FIRST Name: Σinglar LAST Name: Vikter SID (All Digits): 0123456789

Pledge of Academic Integrity

By my honor, I affirm that

(1) this document—which I have produced for the evaluation of my performance—reflects my original, bona fide work, and that I have neither provided to, nor received from, anyone excessive or unreasonable assistance that produces unfair advantage for me or for any of my peers;

(2) as a member of the UC Berkeley community, I have acted with honesty, integrity, respect for others, and professional responsibility—and in a manner consistent with the letter and intent of the campus Code of Student Conduct;

(3) I have not violated—nor aided or abetted anyone else to violate—the instructions for this exam given by the course staff, including, but not limited to, those on the cover page of this document; and

(4) More generally, I have not committed any act that violates—nor aided or abetted anyone else to violate—UC Berkeley, state, or Federal regulations, during this exam.

(10 Points) In the space below, hand-write the following sentence, verbatim. Then write your name in legible letters, sign, include your full SID, and date before submitting your work:

I have read, I understand, and I commit to adhere to the letter and spirit of the pledge above.

I have read, I understand, and I commit to adhere to the letter and spirit of the pledge above.

Full Name: Σinglar Vikter Signature: Σ V

Date: 15 May 2026 Student ID: 0123456789

FIRST Name: Singlar LAST Name: Vikter SID (All Digits): 0123456789

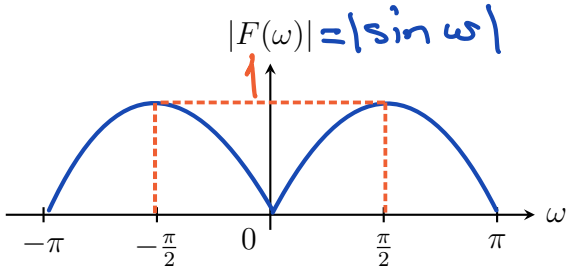
E3.1 (55 Points) Discrete-Time System Analysis

Consider a discrete-time linear time-invariant (DT-LTI) system F whose frequency response is given by

$$F(\omega) = i\sin(\omega)$$

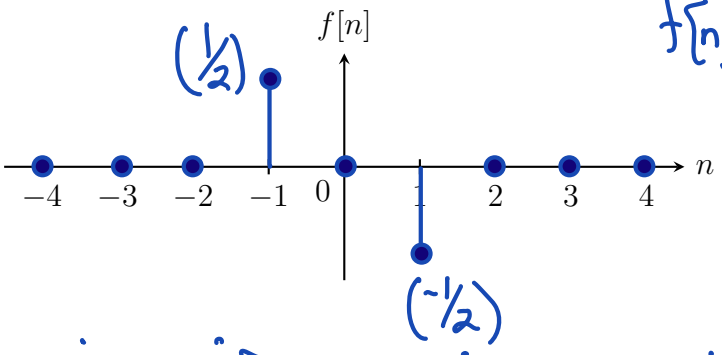
(a) (10 Points) Provide a well-labeled plot of the magnitude response $|F(\omega)|$ for the principal period $\omega \in [-\pi, \pi]$ on the axes provided below.

Based on your plot, determine whether the filter is low-pass, band-pass, or high-pass. Provide a succinct, yet clear and convincing, explanation.



Band-pass

(b) (10 Points) Determine the impulse response $f[n]$ of the system F, and provide a well-labeled stem plot of $f[n]$ on the axes provided below.



$$f[n] = \begin{cases} \frac{1}{2} & n=-1 \\ -\frac{1}{2} & n=1 \\ 0 & \text{e/w} \end{cases}$$

$$F(\omega) = i\sin\omega = \frac{i}{2i}\left(e^{i\omega} - e^{-i\omega}\right) = \frac{1}{2}e^{i\omega} - \frac{1}{2}e^{-i\omega}$$

Recall $F(\omega) = \sum_{n=-\infty}^{\infty} f[n]e^{-i\omega n} \Rightarrow$ $f[-1]$ $f[1]$

FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.1 (Continued)

(c) (15 Points) We apply the following input signal to the system:

$$x[n] = 1 + (-1)^n + \sin\left(\frac{\pi}{2}n\right)$$

Determine a reasonably simple expression for the corresponding output values $y[n]$.

$x[n] = 1 + (-1)^n + \sin\left(\frac{\pi}{2}n\right) = e^{i0n} + e^{i\pi n} + \frac{1}{2i}e^{i\frac{\pi}{2}n} - \frac{1}{2i}e^{-i\frac{\pi}{2}n}$

$y[n] = \cancel{F(0)}^{0}e^{i0n} + \cancel{F(\pi)}^{0}e^{i\pi n} + \frac{1}{2i}F\left(\frac{\pi}{2}\right)e^{i\frac{\pi}{2}n} + \frac{1}{2i}F\left(-\frac{\pi}{2}\right)e^{-i\frac{\pi}{2}n}$

$F(0) = i\sin 0 = 0$

$F(\pi) = i\sin\pi = 0$

$F\left(\frac{\pi}{2}\right) = i\sin\left(\frac{\pi}{2}\right) = i$

$F\left(-\frac{\pi}{2}\right) = i\sin\left(-\frac{\pi}{2}\right) = -i$

$y[n] = \frac{1}{2}e^{i\frac{\pi}{2}} + \frac{1}{2}e^{-i\frac{\pi}{2}n} = \cos\left(\frac{\pi}{2}n\right) \Longrightarrow$

$x[n] = 1 + (-1)^n + \sin\left(\frac{\pi}{2}n\right) \longrightarrow \boxed{F(\omega)} \longrightarrow y[n] = \cos\left(\frac{\pi}{2}n\right)$

Note: No new frequency has been created. The input frequencies 0 and π are eliminated. The frequencies $\pm\frac{\pi}{2}$ survive but are scaled appropriately.

FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.1 (Continued)

(d) (20 Points) We apply the following impulse train as the input to the system:

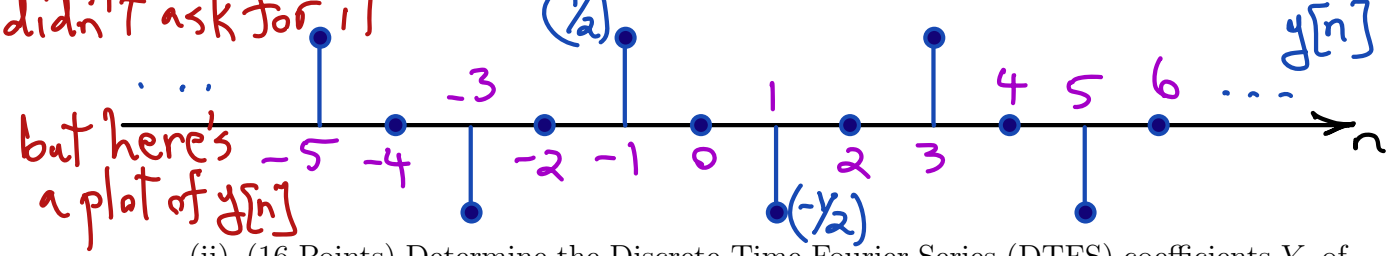
$$x[n] = \sum_{\ell=-\infty}^{\infty} \delta[n - 4\ell]$$

(i) (4 Points) Express the corresponding output $y[n]$ in terms of $f[n]$.

$y[n] = (x * f)[n] = \sum_{\ell=-\infty}^{\infty} f[n-4\ell]$

Convolution of any signal with a shifted impulse shifts the signal to the location of the impulse.

The problem didn't ask for it but here's a plot of $y[n]$



(ii) (16 Points) Determine the Discrete-Time Fourier Series (DTFS) coefficients Y_k of the output $y[n]$ for $k = -1, 0, 1,$ and 2.

$x[n] = \sum_{k\in\langle p\rangle} X_k e^{ik\omega_0 n} \Rightarrow y[n] = \sum_{k\in\langle p\rangle} \underbrace{X_k F(k\omega_0)}_{Y_k} e^{ik\omega_0 n}$

By the eigenfunction property of complex exponentials wrt DT-LTI systems

x is a Uniform 4-periodic Impulse Train

$p=4 \Rightarrow \omega_0 = \frac{2\pi}{p} = \frac{\pi}{2}$

Choose: $\langle p\rangle = \langle 4\rangle = [1,2]$

Recall: $F(\omega) = i\sin\omega$

$x[n] = \frac{1}{4}\sum_{k=-1}^{2} e^{ik\frac{\pi}{2}n}$

$X_k = \frac{1}{p}$

Output Coefficients

$Y_{-1} = X_{-1} F(-\omega_0) = \frac{1}{4} F(-\frac{\pi}{2}) = \frac{-i}{4}$

$Y_0 = X_0 F(0\omega_0) = \frac{1}{4} F(0) = 0$

$Y_1 = X_1 F(\omega_0) = \frac{1}{4} F(\frac{\pi}{2}) = \frac{i}{4}$

$Y_2 = X_2 F(2\omega_0) = \frac{1}{4} F(\pi) = 0$

This was not asked for in the problem but we can derive an analytic expression for the output corresponding to the 4-periodic uniform impulse train $x[n] = \sum_{l=-\infty}^{\infty} \delta[n-4l]$:

$$y[n] = \frac{1}{4}\left(\cancel{F\left(-\tfrac{\pi}{2}\right)e^{-i\frac{\pi}{2}n}}^{-i} + \cancel{F(0)e^{i0n}}^{0} + \cancel{F\left(\tfrac{\pi}{2}\right)e^{i\frac{\pi}{2}n}}^{i} + \cancel{F(\pi)e^{i\pi n}}^{0} \right)$$

$$= \frac{i}{4}\left(e^{i\frac{\pi}{2}n} - e^{-i\frac{\pi}{2}n}\right) = \frac{\cancel{-1}}{\cancel{2i}}\left[\underbrace{\frac{e^{i\frac{\pi}{2}n} - e^{-i\frac{\pi}{2}n}}{2i}}_{\sin\left(\frac{\pi}{2}n\right)}\right] = \frac{-1}{2}\sin\left(\frac{\pi}{2}n\right)$$

$$\boxed{y[n] = \frac{-1}{2}\sin\left(\frac{\pi}{2}n\right) \qquad \forall n \in \mathbb{Z}}$$

$$y[-1] = \frac{-1}{2}\sin\left(-\frac{\pi}{2}\right) = \frac{1}{2}$$

$$y[0] = 0$$

$$y[1] = \frac{-1}{2}\sin\left(\frac{\pi}{2}\right) = \frac{-1}{2}$$

$$y[2] = \frac{-1}{2}\sin\pi = 0$$

These values match the plot of $y[n]$ in Part (d)(i).

FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.2 (65 Points) The Trailing Moving Average Filter

Consider a continuous-time linear, time-invariant (CT-LTI) system H that computes the trailing moving average of its input over a window of length $T > 0$ seconds. The input-output relationship is governed by the following integral equation:

$$y(t) = \frac{1}{T}\int_{t-T}^{t} x(\tau)d\tau \ ,$$

where x and y denote the input and output signals, respectively.

(a) (5 Points) Is the system causal? Provide a succinct, yet clear and convincing, explanation.

Method (I): The word "trailing" in the problem statement hints strongly that the system is causal. To compute $y(t)$, the system uses values of x only in the interval $[t-T, t]$, where $T>0$. So, no "future" value of x is used. Hence, H is causal.

Method II: This can also be established by verifying that the impulse response $h(t)=0, \forall t<0$. Part (c) deals with $h(t)$ in detail. Here, briefly, we note that $h(t)=\frac{1}{T}\int_{t-T}^{t}\delta(\tau)d\tau$. If $t<0$, the interval of integration is entirely to the left of where $\delta(\tau)$ is active $\Rightarrow h(t)=0 \ \forall t<0$

(b) (5 Points) Is the system BIBO stable? Provide a succinct, yet clear and convincing, explanation.

Yes, the system is BIBO stable.

Method I: Consider a bounded input x: $|x(t)| \leq B_x < \infty, \forall t \in \mathbb{R}$.

Then

$$|y(t)| = \left|\frac{1}{T}\int_{t-T}^{t} x(\tau)d\tau\right| \leq \frac{1}{T}\int_{t-T}^{t}|x(\tau)|\,d\tau \leq \frac{1}{T}\int_{t-T}^{t} B_x d\tau = \frac{B_x}{T}\int_{t-T}^{t} d\tau = \frac{B_x}{T}T = B_x$$

So $|y(t)| \leq B_x < \infty$, which means the output is bounded.

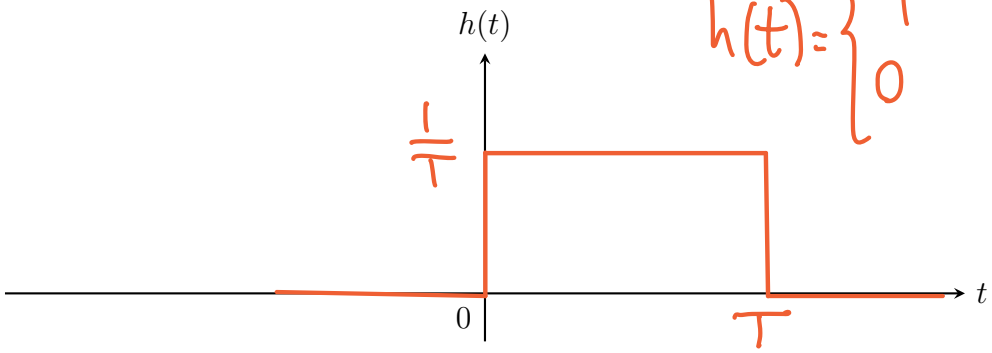
Method II: We can show $\int_{-\infty}^{\infty}|h(t)|\,dt < \infty$. This follows from the fact that H is an FIR filter. The output takes values of the input over a finite interval of duration T seconds. Part (c) explores the exact shape of $h(t)$.

FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.2 (Continued)

(c) (10 Points) Determine a reasonably simple expression for the impulse response $h(t)$ of the system expressed in terms of the CT unit-step $u(t)$.

Provide a well-labeled plot of $h(t)$.



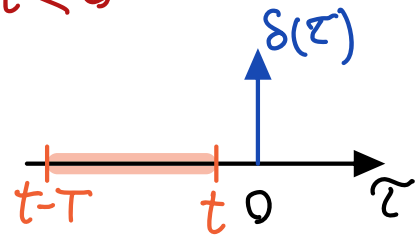
$$h(t) = \begin{cases} \frac{1}{T} & 0 \le t \le T \\ 0 & \text{e/w} \end{cases}$$

$$y(t) = \frac{1}{T}\int_{t-T}^{t} x(\tau)\,d\tau \quad \Rightarrow \quad h(t) = \frac{1}{T}\int_{t-T}^{t} \delta(\tau)\,d\tau$$

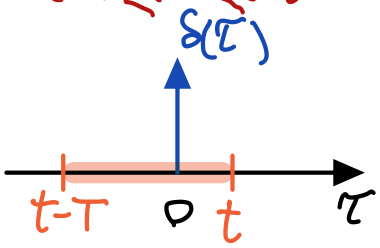
Let $x(\tau) = \delta(\tau)$

Three qualitatively different cases are shown below

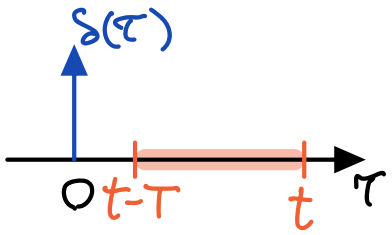
$t < 0$



$t - T \le 0 \le t$



$0 < t - T < t$



(I) $0 \notin [t-T, t]$ (II) $0 \in [t-T, t]$ (III) $0 \notin [t-T, t]$

Case (II) — that is, $t - T \le 0 \le t$ — is the only one where the interval of integration captures the Dirac impulse at the origin $\tau = 0$. The inequalities $t - T \le 0 \le t$ can be rewritten as

$$\begin{cases} 0 \le t \\ \text{and} \\ t - T \le 0 \end{cases} \iff \begin{cases} 0 \le t \\ \text{and} \\ t \le T \end{cases} \iff 0 \le t \le T.$$

For this range of t, $\int_{t-T}^{t} \delta(\tau)\,d\tau = 1 \Rightarrow h(t) = \frac{1}{T}$

FIRST Name: Zinglar LAST Name: Vikter SID (All Digits): 0123456789

E3.2 (Continued)

(d) (10 Points) Show that the frequency response $H(\omega)$ of the system can be expressed in the form $H(\omega) = A(\omega)e^{i\theta(\omega)}$, where the amplitude function is given by

$$A(\omega) = \frac{\sin(\omega T/2)}{\omega T/2}.$$

Furthermore, determine a reasonably simple expression for the phase term $\theta(\omega)$.

Method I: Apply the eigenfunction property of $e^{i\omega t}$

Let $x(t) = e^{i\omega t} \Rightarrow y(t) = H(\omega)e^{i\omega t}$

But $y(t) = \frac{1}{T}\int_{t-T}^{t} x(\tau)\,d\tau = \frac{1}{T}\int_{t-T}^{t} e^{i\omega\tau}\,d\tau = \frac{1}{T}\frac{e^{i\omega\tau}}{i\omega}\Big|_{t-T}^{t}$

$$y(t) = \frac{e^{i\omega t} - e^{i\omega(t-T)}}{i\omega T} = \underbrace{\frac{1 - e^{-i\omega T}}{i\omega T}}_{H(\omega)} e^{i\omega t} \Rightarrow$$

$$H(\omega) = \frac{1 - e^{-i\omega T}}{i\omega T} = \frac{e^{i\omega T/2} - e^{-i\omega T/2}}{i\omega T} e^{-i\omega T/2}$$

Technique of Balancing the Exponents

$$= \frac{2}{\omega T} \underbrace{\frac{e^{i\omega T/2} - e^{-i\omega T/2}}{2i}}_{\sin(\omega T/2)} e^{-i\omega T/2}$$

$\theta(\omega)$: linear phase filter

$$H(\omega) = \underbrace{\frac{\sin(\omega T/2)}{\omega T/2}}_{A(\omega)} e^{-i\omega T/2} \Rightarrow A(\omega) = \frac{\sin(\omega T/2)}{\omega T/2}$$

$$\Rightarrow \theta(\omega) = -\omega T/2$$

E3.2(d) (Continued)

Method II: Derive the frequency response from the impulse response

$$H(\omega) = \int_{-\infty}^{\infty} h(t) e^{-i\omega t} dt \qquad h(t) = \begin{cases} \frac{1}{T} & t \in [0,T] \\ 0 & t \notin [0,T] \end{cases}$$

$$= \int_0^T \frac{1}{T} e^{-i\omega t} dt$$

$$= \frac{1}{T} \int_0^T e^{-i\omega t} dt$$

$$= \frac{1}{T} \frac{e^{-i\omega t}}{-i\omega} \Bigg|_0^T = \frac{1}{T} \frac{e^{-i\omega t}}{i\omega} \Bigg|_T^0$$

$$= \frac{e^{-i\omega 0} - e^{-i\omega T}}{i\omega T} = \frac{1}{\omega T} \frac{1 - e^{-i\omega T}}{i}$$

$$= \frac{2}{\omega T} \underbrace{\frac{e^{i\omega T/2} - e^{-i\omega T/2}}{2i}}_{\sin(\omega T/2)} e^{-i\omega T/2}$$

$$H(\omega) = \underbrace{\frac{\sin(\omega T/2)}{\omega T/2}}_{A(\omega)} \underbrace{e^{-i\omega T/2}}_{e^{i\theta(\omega)}}$$

$$\Rightarrow A(\omega) = \frac{\sin(\omega T/2)}{\omega T/2}$$

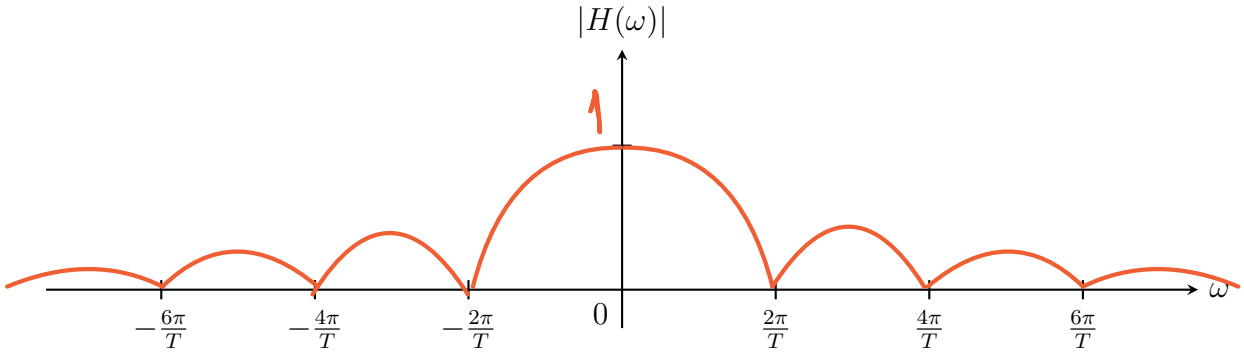
$$\Rightarrow \theta(\omega) = -\omega T/2$$

FIRST Name: Zinglar LAST Name: Vikter SID (All Digits): 0123456789

E3.2 (Continued)

(e) (10 Points) Provide a well-labeled plot of the magnitude response $|H(\omega)|$ over the frequency range $\omega \in \left[-\frac{7\pi}{T}, \frac{7\pi}{T}\right]$.

Be sure to identify the "null frequencies" (the zero crossings of the frequency response).



$|H(\omega)| = |A(\omega)e^{i\theta(\omega)}| = |A(\omega)| \Rightarrow$

$|H(\omega)| = \left|\frac{\sin(\omega T/2)}{\omega T/2}\right|$

By L'Hopital's Rule or by recognizing that since $H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-i\omega t}dt$, we have $H(0) = \int_{-\infty}^{\infty} h(t)dt =$ Area under the impulse response, we conclude that $H(0) = 1 \Rightarrow |H(0)| = 1$.

Behavior of $A(\omega)$ is sinc-like. So, $|H(\omega)|$ is abs val of that. First zero crossing for $\omega > 0$ is at solution of $\frac{\omega T}{2} = \pi \Rightarrow \omega = \frac{2\pi}{T}$. All other zero crossings are at nonzero integer multiples of $\frac{2\pi}{T}$. See the plot above.

E3.2 (Continued)

(f) (10 Points) Suppose the input to the system is

$$\forall t \in \mathbb{R}, \qquad x(t) = \cos^2\left(\frac{\pi}{T}t\right).$$

Determine a reasonably simple expression for the output $y(t)$.

Provide a succinct, yet clear and convincing, intuitive explanation of this result both in the time and frequency domains.

Reliable carrier pigeons have informed us that $\cos^2\theta = \dfrac{1+\cos(2\theta)}{2}$.

$x(t) = \cos^2\left(\frac{\pi}{T}t\right) = \frac{1}{2} + \frac{1}{2}\cos\left(\frac{2\pi}{T}t\right)$

$\cos\left(\frac{2\pi}{T}t\right) = \frac{1}{2}e^{i\frac{2\pi}{T}t} + \frac{1}{2}e^{-i\frac{2\pi}{T}t}$ ← Both are null frequencies of $H(\omega)$. So, they get eliminated by the filter.

The only input frequency that survives is the zero-frequency term $\frac{1}{2}$. Therfore, $y(t) = H(0)\cdot\frac{1}{2} = \frac{1}{2}$

$$y(t) = \frac{1}{2} \quad \forall t \in \mathbb{R}$$

The filter averages the input signal over T seconds. Since $\cos\left(\frac{2\pi}{T}t\right)$ has period T seconds, it averages to zero. Hence, eliminated.

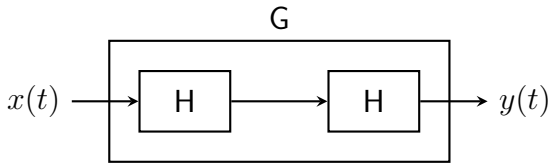
(g) (5 Points) Can the system be implemented using a linear constant coefficient differential equation (LCCDE)? Provide a succinct, yet clear and convincing, explanation.

No. Only a CT-LTI system whose frequency response can be expressed as a rational function of $i\omega$ has an LCCDE implementation. Our filter's frequency response can not be expressed as a polynomial in $i\omega$ over another polynomial in $i\omega$. The sine function and the exponential $e^{i\theta(\omega)}$ don't allow it.

FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.2 (Continued)

(h) (10 Points) We construct a new system G by placing two identical copies of the system H in cascade (series), as shown in the diagram below.



Determine a reasonably simple expression for $|G(\omega)|$, the magnitude response of G.

$g(t) = (h * h)(t) \Rightarrow G(\omega) = H(\omega)H(\omega) = H^2(\omega) \Rightarrow$

$G(\omega) = A^2(\omega)e^{i2\theta(\omega)} \Rightarrow |G(\omega)| = A^2(\omega) \Rightarrow$

$$|G(\omega)| = \frac{\sin^2(\omega T/2)}{(\omega T/2)^2}$$

FIRST Name: Singlar LAST Name: Vikter SID (All Digits): 0123456789

E3.3 (65 Points) SVD, PCA, and Exam Scores

We want to analyze the exam performances of five students in a class. We record their Calculus and Physics scores in a 2×5 data matrix $\mathbf{X}$. The first row contains the students' Calculus scores, and the second row their Physics scores. Each column represents a single student. The raw data matrix that contains the scores is

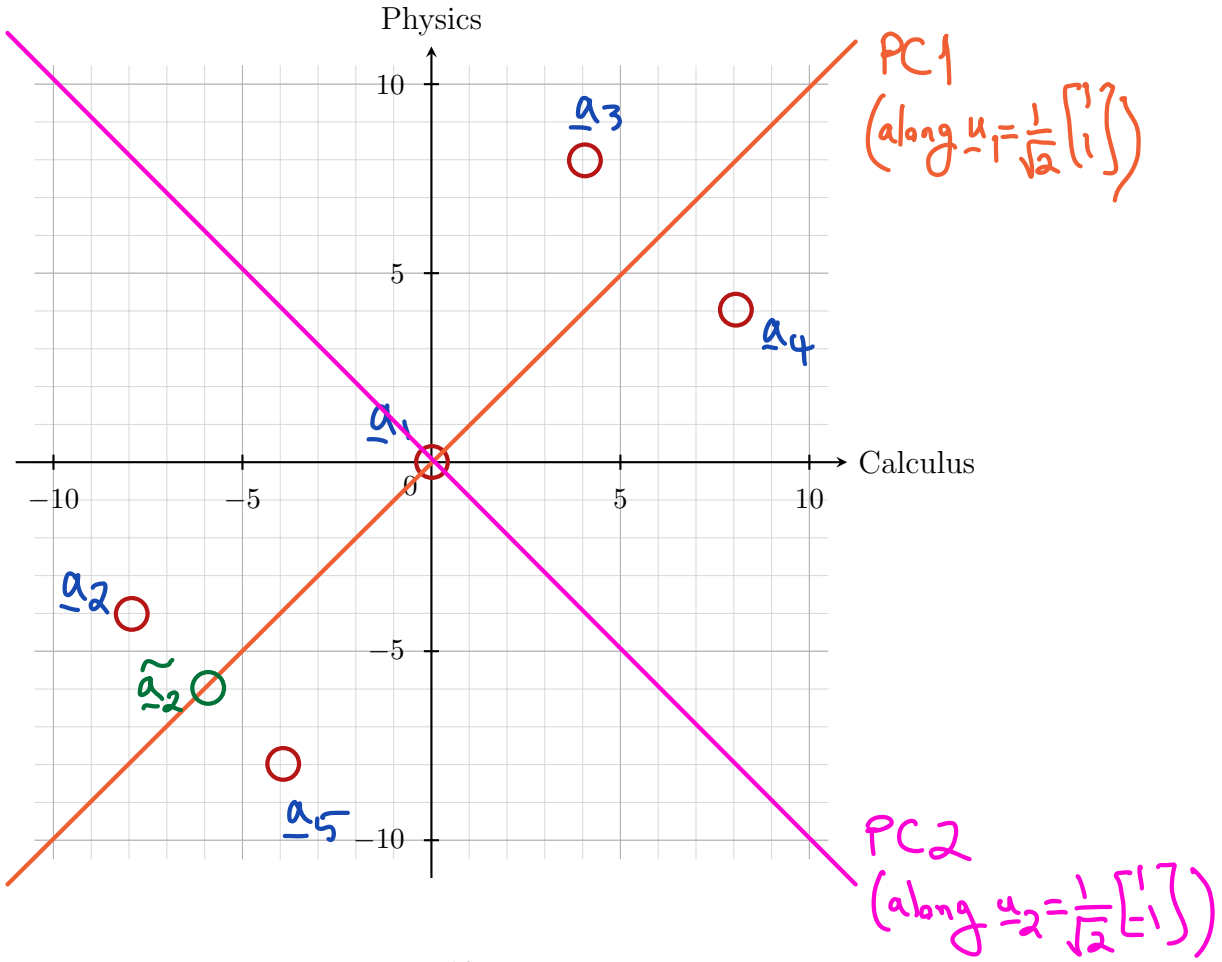
$$\mathbf{X} = \begin{bmatrix} 80 & 72 & 84 & 88 & 76 \\ 75 & 71 & 83 & 79 & 67 \end{bmatrix} = \begin{bmatrix} \underline{x}_1 & \underline{x}_2 & \underline{x}_3 & \underline{x}_4 & \underline{x}_5 \end{bmatrix}$$

(a) (10 Points) To prepare for Principal Component Analysis (PCA), we must first mean-center the data. To do so, we de-mean the *columns* of the raw data matrix $\mathbf{X}$—that is, we subtract the mean of the Calculus scores from the first row of $\mathbf{X}$, and subtract the mean of the Physics scores from the second row of $\mathbf{X}$.

Show that the de-meaned data matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} 0 & -8 & 4 & 8 & -4 \\ 0 & -4 & 8 & 4 & -8 \end{bmatrix} = \begin{bmatrix} \underline{a}_1 & \underline{a}_2 & \underline{a}_3 & \underline{a}_4 & \underline{a}_5 \end{bmatrix}$$

and plot the de-meaned data on the grid below. You may use the blank space at the top of the next page to show your work.



FIRST Name: Einglar LAST Name: Vikter SID (All Digits): 0123456789

E3.3 Part (a) (Continued)

$\in \mathbb{R}^{2\times 5}$ (m, n)

Strictly speaking:

$$\bar{x} = \frac{1}{n} X \mathbb{1} = \frac{1}{5}\begin{bmatrix} 80 & 72 & 84 & 88 & 76 \\ 75 & 71 & 83 & 79 & 67 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 80 \\ 75 \end{bmatrix}$$

But a keen observer would note that the numbers in the top row are symmetrically distributed around 80 — and those in the second row around 75. In effect, this turns Student 1 into an "average student," which is why $\underline{a}_1 = \underline{0}$.

$$A = \begin{bmatrix} \underline{x}_1 - \bar{x} & \underline{x}_2 - \bar{x} & \underline{x}_3 - \bar{x} & \underline{x}_4 - \bar{x} & \underline{x}_5 - \bar{x} \end{bmatrix} =$$

$$\begin{bmatrix} 80-80 & 72-80 & 84-80 & 88-80 & 76-80 \\ 75-75 & 71-75 & 83-75 & 79-75 & 67-75 \end{bmatrix} = \begin{bmatrix} 0 & -8 & 4 & 8 & -4 \\ 0 & -4 & 8 & 4 & -8 \end{bmatrix}$$

(b) (5 Points) To determine the principal components, why is it that we must examine the properties of the matrix $\mathbf{A}\mathbf{A}^\mathsf{T}$—and *not* $\mathbf{A}^\mathsf{T}\mathbf{A}$?

Since the students are pegged to the columns, and the features of each student (i.e., their Calculus & Physics scores) arranged columnwise, it's the left singular vectors $\underline{u}_1, \underline{u}_2$ that we're after (since they span the column space of X). Hence, it's the AA^T matrix that we examine. The eigenvectors of AA^T are the $\underline{u}$ vectors, and the eigenvalues of AA^T are the squares of the singular values.

E3.3 (Continued)

(c) (35 Points) Consider the matrix

$$\mathbf{A}\mathbf{A}^{\mathsf{T}} = \begin{bmatrix} 160 & 128 \\ 128 & 160 \end{bmatrix}.$$

(i) (15 Points) Perform eigenanalysis on $\mathbf{A}\mathbf{A}^{\mathsf{T}}$. In particular, determine its eigenvalue-eigenvector pairs (eigenpairs) $(\lambda_1, \boldsymbol{z}_1)$ and $(\lambda_2, \boldsymbol{z}_2)$, where

$$\mathbf{A}\mathbf{A}^{\mathsf{T}}\boldsymbol{z}_1 = \lambda_1 \boldsymbol{z}_1, \quad \mathbf{A}\mathbf{A}^{\mathsf{T}}\boldsymbol{z}_2 = \lambda_2 \boldsymbol{z}_2, \quad \text{and} \quad \lambda_1 \geq \lambda_2.$$

Method I:

Note that AA^T has the constant row sums $160+128=288$.

So $\mathbb{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector corresponding to eigenvalue 288.

So $(\lambda_1 = 288, \underline{z}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix})$.

Since AA^T is symmetric, the other eigvec must be orthogonal to $\mathbb{1}$. So, we choose $\underline{z}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Multiplying we have:

$$AA^T \underline{z}_2 = \begin{bmatrix} 160 & 128 \\ 128 & 160 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 32 \\ -32 \end{bmatrix} = 32\begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow (\lambda_2 = 32, \underline{z}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix})$$

Method II: Brute force eigendecomposition on the 2x2 matrix AA^T

$$\lambda I - AA^T = \begin{bmatrix} \lambda-160 & -128 \\ -128 & \lambda-160 \end{bmatrix} \qquad |\lambda I - AA^T| = (\lambda-160)^2 - 128^2$$

$$= (\lambda-160-128)(\lambda-160+128)$$

$$= (\lambda-288)(\lambda-32)$$

$\Rightarrow \lambda_1 = 288, \lambda_2 = 32$

$$\lambda_1 I - AA^T = \begin{bmatrix} 128 & -128 \\ -128 & 128 \end{bmatrix} \Rightarrow \underline{z}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$ (since $\lambda_1 I - AA^T$ has row sums equal to zero)

Similarly,

$$\lambda_2 I - AA^T = \begin{bmatrix} -128 & -128 \\ -128 & -128 \end{bmatrix} \Rightarrow \underline{z}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$ (since the two columns of $\lambda_2 I - AA^T$ are identical)

E3.3 (Continued)

(ii) (10 Points) The first left singular pair of the matrix $\mathbf{A}$ is given by

$$\left(\sigma_1 = 12\sqrt{2}, \boldsymbol{u}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\right).$$

Compute the second left singular pair $(\sigma_2, \boldsymbol{u}_2)$.

$\sigma_1 = \sqrt{\lambda_1} = \sqrt{288} = \sqrt{144 \cdot 2} = 12\sqrt{2}$

$\underline{u}_1 = \frac{\underline{z}_1}{\|\underline{z}_1\|} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}$

$\sigma_2 = \sqrt{\lambda_2} = \sqrt{32} = \sqrt{16 \cdot 2} = 4\sqrt{2}$

$\underline{u}_2 = \frac{\underline{z}_2}{\|\underline{z}_2\|} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$

(iii) (10 Points) The left singular vectors $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ represent the principal component directions (PC1 and PC2, respectively) of our data points. On the grid provided for Part (a), draw straight lines across the data points along PC1 and PC2. Explain in plain English terms what feature of the dataset PC1 captures.

PC1 captures the direction of the maximum variance (spread) of the de-meaned data points. We can see from the scatter plot that the data are spread maximally along the 45° "Northeast-Southwest Corridor."

FIRST Name: Singlar LAST Name: Vikter SID (All Digits): 0123456789

(d) (8 Points) Calculate the first right singular vector $\boldsymbol{v}_1$ corresponding to σ_1.

Hint: Do not attempt to determine the eigenvectors of the 5×5 matrix $\mathbf{A}^\mathsf{T}\mathbf{A}$. Use fundamental concepts of SVD and any relevant result from the previous part.

Since $\underline{u}_1$ is a left singular vector, we know that

$$\underline{u}_1^T A = \sigma_1 \underline{v}_1^T \Rightarrow A^T \underline{u}_1 = \sigma_1 \underline{v}_1 \Rightarrow \underline{v}_1 = \frac{1}{\sigma_1} A^T \underline{u}_1 \Rightarrow$$

$$\underline{v}_1 = \frac{1}{12\sqrt{2}} \begin{bmatrix} 0 & 0 \\ -8 & -4 \\ 4 & 8 \\ 8 & 4 \\ -4 & -8 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{12\sqrt{2}\sqrt{2}} \begin{bmatrix} 0+0 \\ -8-4 \\ 4+8 \\ 8+4 \\ -4-8 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 0 \\ -12 \\ 12 \\ 12 \\ -12 \end{bmatrix} \Rightarrow$$

$$\underline{v}_1 = \frac{1}{2} \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

(e) (7 Points) We want to approximate the scores of Student 2 using only the first principal component direction $\boldsymbol{u}_1$. Let $\boldsymbol{a}_2$ denote the second column of the de-meaned matrix $\mathbf{A}$—representing the de-meaned scores of Student 2.

Let $\widetilde{\boldsymbol{a}}_2 = \alpha_2 \boldsymbol{u}_1$—where α_2 is an appropriately-chosen scalar—denote the approximation of $\boldsymbol{a}_2$ along the first principal component direction. Determine the projection of $\boldsymbol{a}_2$ onto the first principal component direction $\boldsymbol{u}_1$, and from there calculate $\widetilde{\boldsymbol{a}}_2$.

$$\underline{a}_2 = \alpha_2 \underline{u}_1 + \tilde{\alpha}_2 \underline{u}_2 \cdot \qquad \underline{\tilde{a}}_2 = \alpha_2 \underline{u}_1$$

$$\alpha_2 = \frac{\langle \underline{a}_2, \underline{u}_1 \rangle}{\langle \underline{u}_1, \underline{u}_1 \rangle} = \langle \underline{a}_2, \underline{u}_1 \rangle = \underline{a}_2^T \underline{u}_1 = \begin{bmatrix} -8 & -4 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \alpha_2 = \frac{1}{\sqrt{2}}(-12) = \frac{-12}{\sqrt{2}} = -6\sqrt{2} \Rightarrow$$

$$\underline{\tilde{a}}_2 = -6\sqrt{2}\, \underline{u}_1 = -6\sqrt{2} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -6 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ -6 \end{bmatrix}$$

Marked in green on the scatterplot.